

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

MARK SCHEME

CLASS

INDEX NUMBER

BIOLOGY

9648/02

Paper 2 Core Paper

19 August 2016

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	20
2	11
3	12
4	12
5	13
6	12
Section B	
7 / 8	20
Total	100

This document consists of **18** printed pages.



Section AAnswer **all** questions.**1** A protease is an enzyme that digests protein.**(a)** With the aid of a diagram, describe the reaction catalysed by protease.

diagram showing substrates – polypeptide / dipeptide and water
breaking of peptide bond
products – amino acids
with appropriate labels;

1. hydrolysis reaction

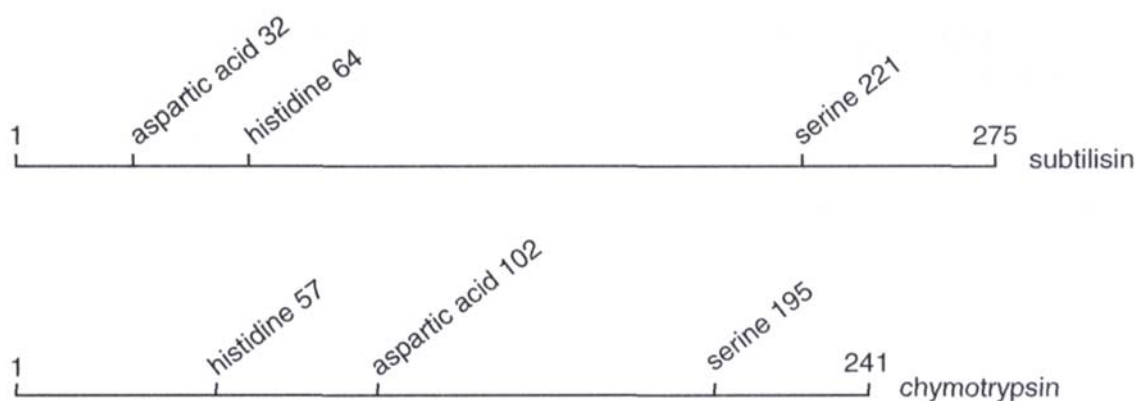
with the addition of a water molecule;

2. breaking of peptide bond (-CONH)

btwn amino acids;

[3]

Subtilisin is a protease synthesised by bacteria and is made up of 275 amino acids. Chymotrypsin is a protease synthesised by bacteria and is made up of 241 amino acids. Both enzymes have the same arrangement of three amino acids, serine, histidine and aspartic acid in their active sites but they are structurally different with the three amino acids being in different positions in the amino acid sequences shown in Fig. 1.1.

**Fig. 1.1**

- (b) With reference to Fig. 1.1 and your knowledge on levels of protein structure, describe how amino acid residues at different positions in the protein may be brought together in the active site.

1. 1° str is the unique linear seq of aa

e.g. aspartic acid at position 32, histidine at 64 and serine at 221;

2. folding of the polypeptide to form 3° str held by R group interactions (e.g. hydrophobic int, H bond, ionic bond, disulfide bridges)

to bring together catalytic and contact residues in the active site ➡ specific 3D config;

[2]

The graph in Fig. 1.2 shows how the activity of a protease varies with temperature.

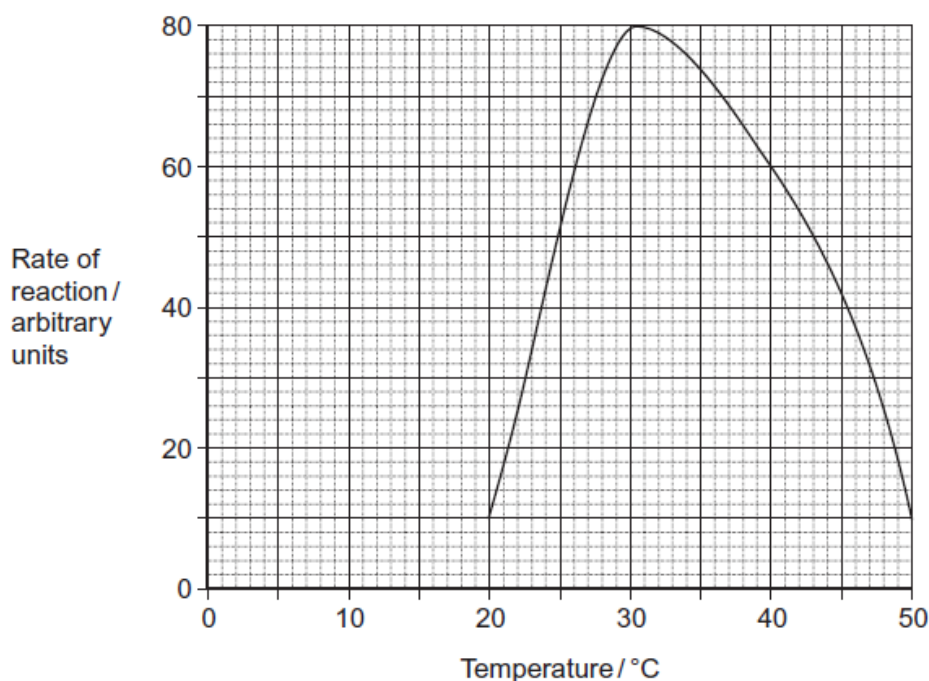


Fig. 1.2

- (c) With reference to Fig. 1.2, explain the shape of the graph between 30°C and 50°C.

1. sharp ↓ in rate of reaction from 30 to 50 °C

from 80 to 10 au;

2. further ↑ in temp beyond opt temp ➡ ↑ in KE of E

violent molecular vibrations ➡ breaking of intramolecular H bonds and hydrophobic interactions;

3. denaturation of enz due to unfolding of enz / disrupt 3° str

leading loss of specific 3D config of active site ➡ E no longer able to bind to S;

[3]

Students investigated the effect of pH on the activity of protease.

- The students used agar plates containing protein. The protein made the agar cloudy.
- They made four wells of equal size in the agar of each plate.
- They added a drop of protease solution to each of the wells. The protease solution in each well was at different pH.
- The students incubated the agar plates for 4 hours at constant temperature.

Fig. 1.3 shows the agar plates after they were incubated and the pH of the protease solution in each well.

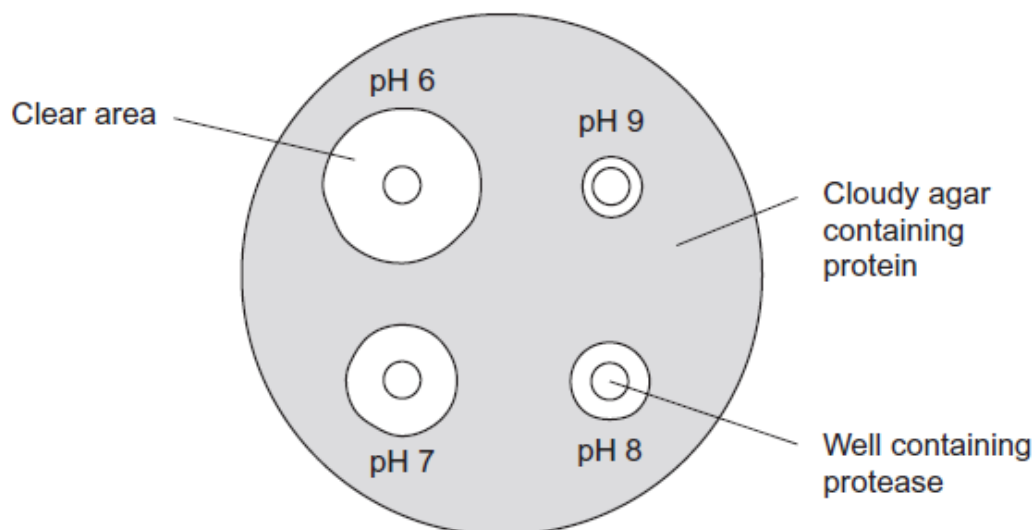


Fig. 1.3

(d) With reference to Fig. 1.3,

- (i) suggest a suitable temperature for incubating the agar plates. Explain your answer.

1. **30°C;**

2. **opt temp of enz**

highest rate of rxn / temp is not limiting;

[2]

- (ii) explain the effect of pH on the activity of this protease.

1. **opt pH at pH 6 as seen by the largest clear area ➡ highest rate of digestion of protein in agar plate by protease;**

2. **↑ in pH ➡ ↓ in enz activity as seen by ↓ in size of clear area;**

[1]

The gene encoding for protease is transcribed to give rise to mRNA. The mRNA transcribed leaves the nucleus and is translated by ribosomes in the cytosol. Fig. 1.4 shows the process of translation.

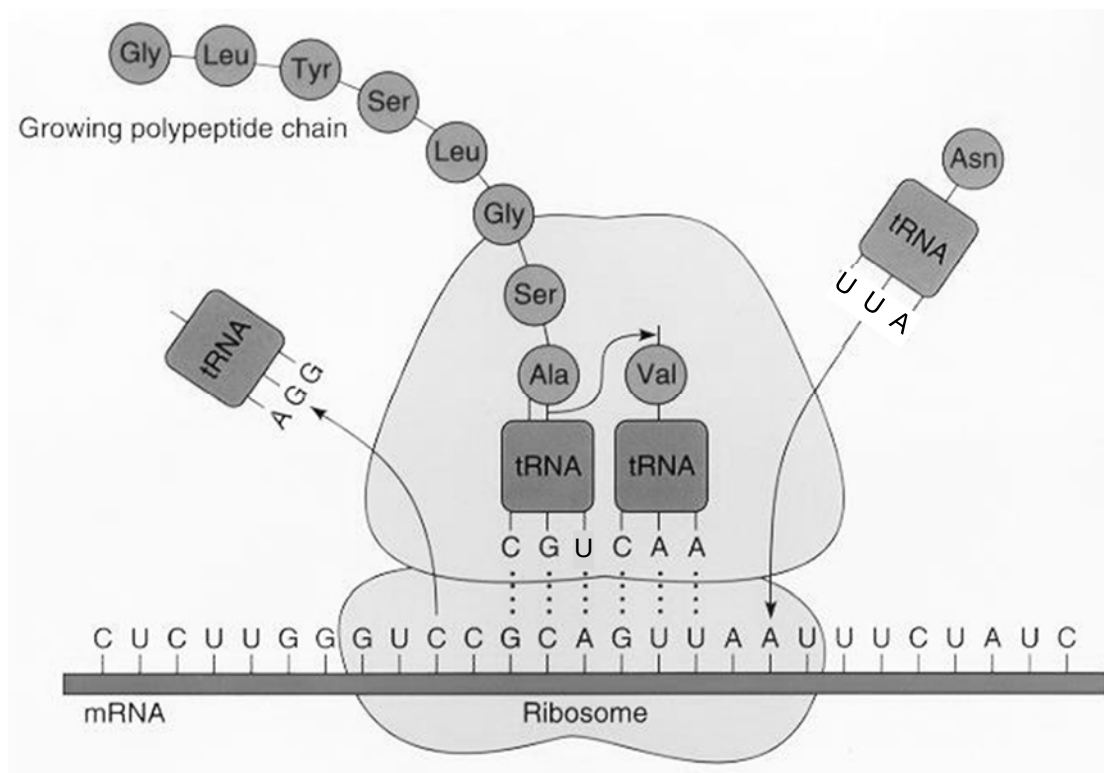


Fig. 1.4

- (e) With reference to Fig. 1.4, describe how the amino acid asparagine (Asn) is incorporated into the growing polypeptide chain.

1. tRNA with anticodon UUA carrying the aa asn

binds to mRNA codon AAU via formation of H bonds btwn comp bp at A site of large ribosomal subunit;

2. formation of peptide bond between asn and val

catalysed by peptidyl transferase;

3. ribosome translocates in 5' to 3' direction

tRNA with (asn and) growing polypeptide chain is now at P site;

4. tRNA carrying val / with anticodon CAA is now at the E site

and is released;

[4]

Aminoacyl tRNA synthetase is an enzyme that catalyses the attachment of amino acids to its corresponding tRNA.

(f) Explain how the structure of tRNA is adapted to perform its function.

1. contains amino acid attachment arm

for attachment of specific amino acid (via ester bond);

2. contains anticodon

that binds to mRNA codon via H bonds btwn comp bp;

3. clover-leaf shape

contributes to specific 3D config ➡ allows binding to ribosome aminoacyl tRNA synthetase; [2]

(max 2m)

(g) A specific aminoacyl tRNA synthetase allows the amino acid asparagine to bind to its active site. A single base substitution occurred in the gene coding for this enzyme, resulting in the enzyme recognising proline instead of asparagine.

Fig. 1.5 shows the effect of this mutation on the resultant aminoacyl tRNA.

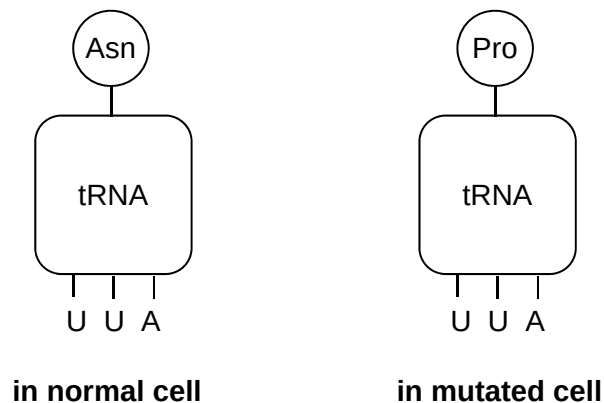


Fig. 1.5

Suggest why a mutation in this gene would have extremely damaging effects on the organism.

1. enz would attach pro instead of asn

to the tRNA with anticodon for asn / UUA;

2. during translation, mRNA codon AAU will code for pro instead of asn

all resultant polypeptides will have asn replaced by pro;

3. Δ in R group of all resultant polypeptides ➡ Δ in R group interactions

Δ in specific 3D config of all resultant prot ➡ affect cellular f(x); [3]

[Total: 20]

2 HIV is a retrovirus that causes acquired immune deficiency syndrome (AIDS).

(a) Outline the process by which HIV enters the host cell.

1. ***gp 120 (and gp 41) on HIV envelope***

recog and binds to CD4 cell surface receptors on T cells;

2. ***viral env fuses with CSM of host cell***

releases viral nucleocapsid into host cell cytoplasm;

[2]

Many integrase inhibitors have been discovered in recent years, and some of them are presently in clinical trials. One such inhibitor is Raltegravir.

In a clinical trial, one group of patients was given Raltegravir while the control group was given a placebo (drug-free pills) over 24 weeks. The number of copies of HIV-1 RNA per ml of blood plasma was measured and the results are given in Fig. 2.1.

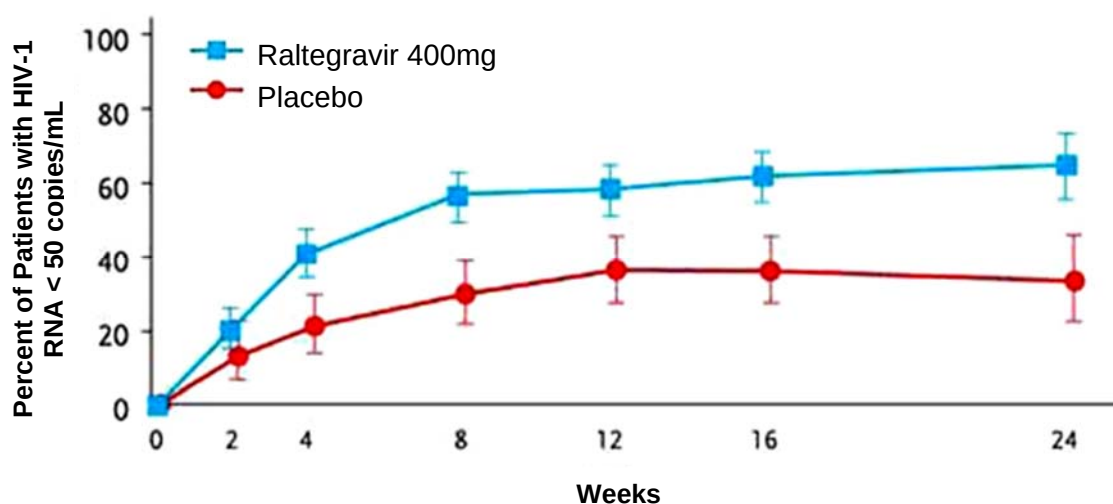


Fig 2.1

(b) With reference to Fig. 2.1, describe the difference in the effect of treating patients with Raltegravir and placebo.

1. ***higher percentage of patients with HIV-1 RNA<50 copies per ml in grp treated with Raltegravir vs placebo throughout 24 week treatment;***

2. ***[QV] for Raltegravir grp: ↑ to 60% patients with HIV-1 RNA< 50 copies at week 8***

for placebo grp: ↑ to 30% patients with HIV-1 RNA< 50 copies at week 8;

[2]

(c) Suggest how Raltegravir works as an antiretroviral drug.

1. *(inhibits integrase) ➔ inhibits integration of HIV DNA into human chr*

preventing formation of provirus;

2. *resulting in HIV unable to undergo replication / viral protein synthesis*

preventing formation of new virions;

[2]

(d) The viral genome undergoes frequent mutation.

Suggest why this means that the action of Raltegravir may no longer be effective in future.

1. *mutations in viral genome alter gene seq coding for integrase*

results in change in 3D conformation of enzyme;

2. *inability of Raltegravir to bind / inhibit integrase*

due to non-complementary to binding site / active site;

[2]

Another important process in the reproductive cycle of HIV is the cleaving of polypeptides into functional proteins using protease. Fig. 2.2 shows such a process.

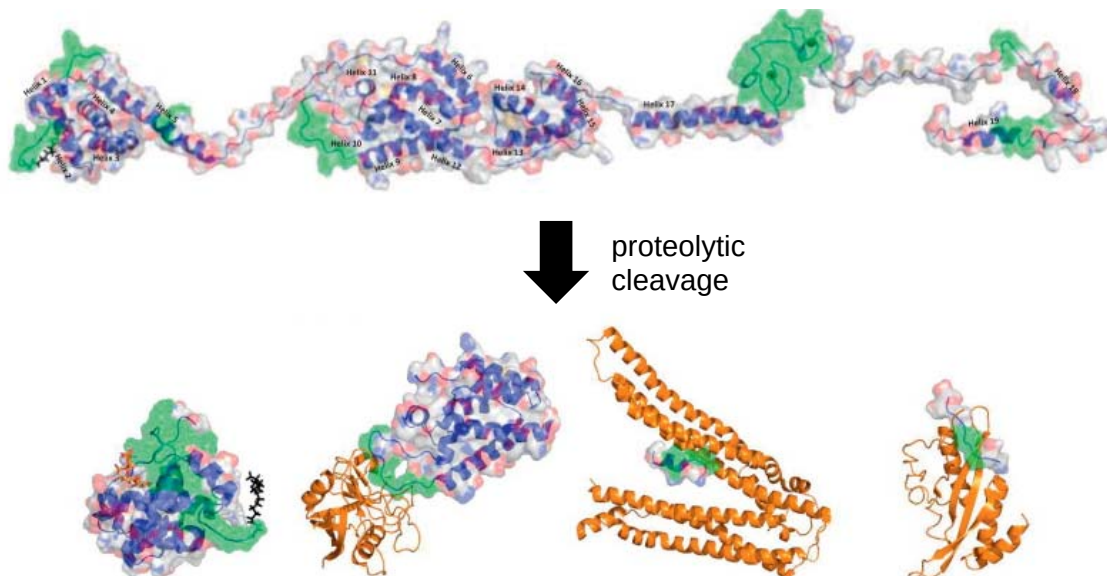


Fig 2.2

(e) Describe the polyproteins and the products formed after proteolytic cleavage.

1. Pol polyprotein cleaved by viral protease

produces enzymes such as RT / IN / HIV protease;

2. Gag polyprotein cleaved by viral protease

produces structural proteins such as capsid proteins;

3. Env polyprotein cleaved by cellular protease

produces glycoproteins gp120 and gp41;

[3]

[Total: 11]

3 Fig. 3.1 shows the structure of a nucleosome.

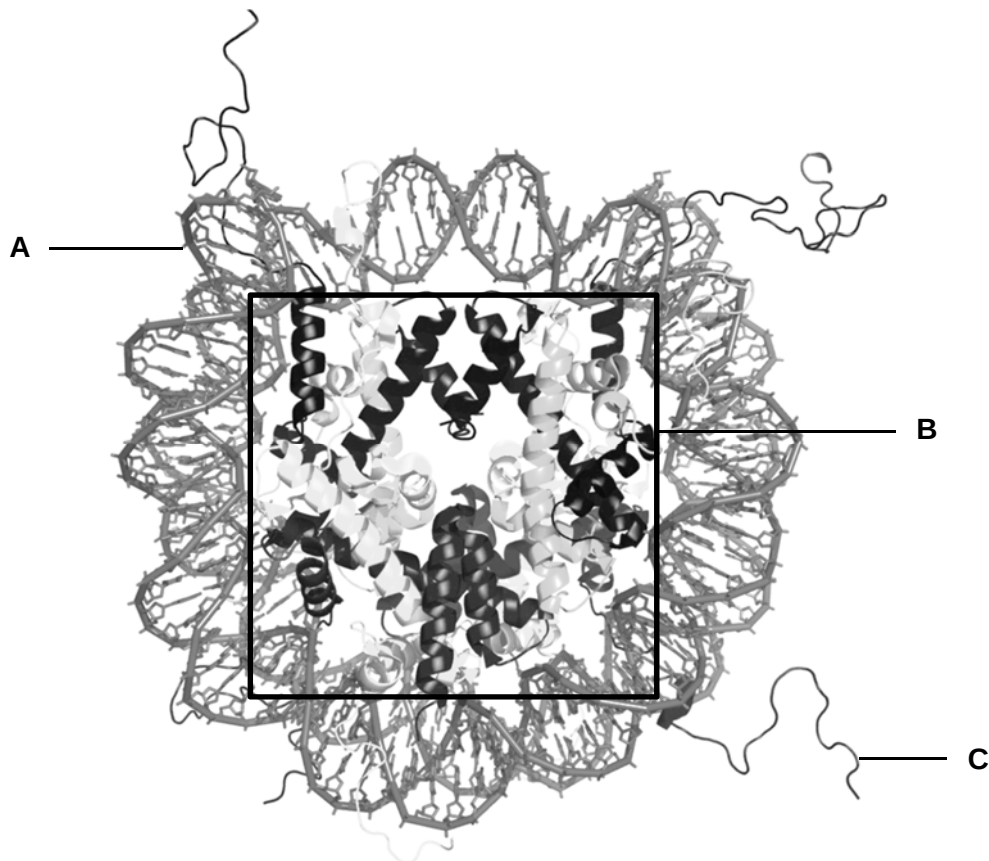


Fig. 3.1

(a) With reference to Fig. 3.1,

(i) label the structures A to C.

A *deoxyribonucleic acid*

B *histone protein / octamer*

C *histone tail;;*

[2]

(2m for all correct ans, -1m for any wrong ans)

(ii) describe how structure **A** is stabilised.

1. **hydrogen bonds btwn comp bp**

where $A = T$, $C \equiv G$;

2. **hydrophobic interactions**

btwn stacked bases;

[2]

(iii) state the secondary structure observed in structure **B**.

α -helix;

[1]

(iv) describe how structure **C** can be modified to influence gene expression.

1. **acetylation of (lys residues on) histone tails by histone acetyltransferase**

$\downarrow$ +ve charge of histones $\Rightarrow$ $\downarrow$ histone's affinity for -ve charged DNA;

2. **chromatin in more relaxed conformation**

allow access of TFs and RNA pol to promoter $\Rightarrow$ $\uparrow$ transcription;

[2]

(accept rev argument, i.e. deacetylation)

(b) Describe how DNA is packaged in eukaryotes.

1. **DNA molecule wrapped $1\frac{3}{4}$ times (146bp) around histone octamer, forming a nucleosome core;**

2. **nucleosome core assoc with H1 prot forming the complete nucleosome;**

3. **linked by spacer DNA forming 10nm 'beads on a string' structure;**

4. **further coiled into solenoid with 6 nucleosomes per turn, forming 30nm fibre;**

5. **30 nm fibre assoc with scaffold prot $\Rightarrow$ 300 nm fiber;**

6. **further coiling to form 700 nm chromatid;**

[5]

(max 5m)

[Total: 12]

- 4 (a) Explain what is meant by *epistasis*.

1. *interaction between two genes*

coding for gene pdts controlling the same characteristic / in same metabolic pathway;

2. *epistatic gene overrides expression of hypostatic gene*

to express its own phenotype;

[2]

In some species of plants, malvidin is a primary flower pigment that results in the colour wine red. Gene **A/a** codes for an enzyme that synthesises malvidin while gene **H/h** was discovered to inhibit its synthesis resulting in white flowers.

In one experiment, double heterozygote parents were crossed to produce F1 generation progeny comprising of 452 white flowering plants and 114 red wine flowering plants.

- (b) Using the symbols for the alleles stated above, draw a genetic diagram to show the expected phenotypic ratio for the F1 offspring.

parental phenotype
(1m)

white flower
plant

x

white flower
plant

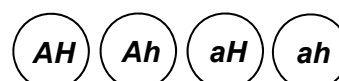
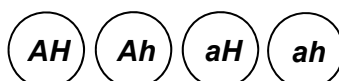
parental genotype

AaHh

x

AaHh

gametes
(1m)



punnett square (1m)

Gametes	AH	Ah	aH	ah
AH	AAHH	AAHh	AaHH	AaHh
Ah	AAHh	AAhh	AaHh	Aahh
aH	AaHH	AaHh	aaHH	aaHh
ah	AaHh	Aahh	aaHh	aahh

offspring genotype
(1m)

$9 \text{ A_H_} : 3 \text{ aaH_} : 1 \text{ aahh} : 3 \text{ A_hh}$

offspring phenotypic
ratio (1m) (*genotype must correlate with phenotype*)

13 white

: 3 wine red

[5]

- (c) Determine, using an appropriate statistical test, if the observed results are expected. The critical values for the chi-squared test are provided in Table 4.1.

Table 4.1

degree of freedom	Probability, p				
	0.1	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Let H_0 be the inheritance of plant colour that follows the ratio of 13:3

Phenotype	Expected ratio	Observed (O)	Expected (E)	$\frac{(O - E)^2}{E}$
white	13	452	460	0.1349
red	3	114	106	0.5844
df = 2 - 1	Total =	566		

Hence, $\chi^2_{(calculated)} = 0.719$ (3s.f.)

1. correct working;
2. correct answer;
3. at $p = 0.05$, $df = 1$, critical $\chi^2 = 3.84$
calculated $\chi^2 < \text{critical } \chi^2$;
4. deviation btw observed and expected ratios is not statistically significant and the difference is due to chance alone;
5. hence, H_0 is not rejected, inheritance of plant colour follows the expected ratio of 13:3;

ECF for 12:4 and 9:7 in (b)

For 12:4

(E) 424.5, 142.5

$\chi^2 = 71.2$, reject H_0

For 9:7

(E) 318, 248

$\chi^2 = 128.9$, reject H_0

[5]

[Total: 12]

- 5 (a) A myelinated axon transmits impulses faster than a non-myelinated axon.

Explain this difference.

1. in myelinated neurone, myelin (is lipid-rich thus) act as electrical insulator

preventing movement of ions across axolemma;

2. cause AP to be generated only at nodes of Ranvier / nerve impulse 'jumps' from node to node

via saltatory conduction;

3. in non-myelinated axon, AP is generated along whole length of axon

via continuous conduction;

[2]

Doctors investigated the relationship between myelin in the brain tissue and different types of dementia. All types of dementia involve loss of mental ability.

The doctors measured the mean amount of myelin in samples of brain tissue from:

- a control group of 12 people without dementia
- 20 people with vascular dementia (VaD)
- 19 people with Alzheimer's dementia (AD)
- 31 people with Lewy body dementia (LD).

The doctors' results are shown in Fig. 3.1. The vertical bars show standard errors.

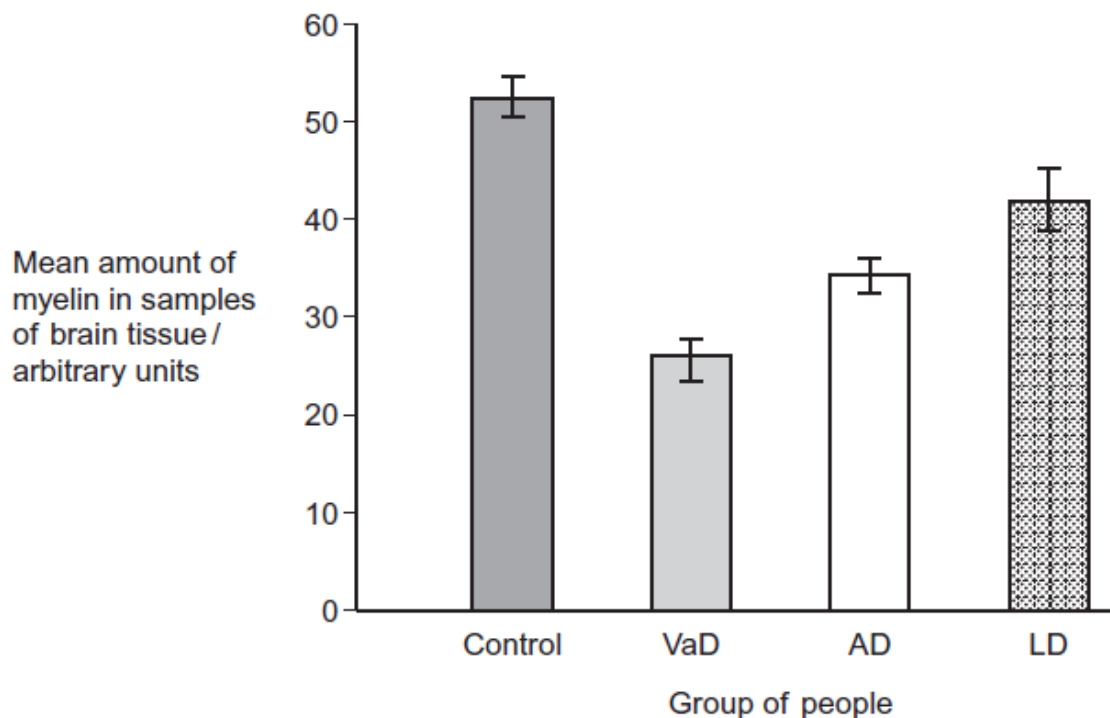


Fig. 5.1

(b) With reference to Fig. 5.1,

(i) explain the significance of the standard error bars.

1. **to indicate the range of amount of myelin in samples of brain tissue**

to allow for statistical comparison;

[1]

(ii) discuss whether the data supports the conclusion that there is a relationship between incidence of dementia and the amount of myelin in a person's brain.

1. **mean amt of myelin lower in all dementia groups cf control**

[QV] control: 52 au, VaD: 25 au, AD: 33 au, LD: 41 au;

2. **[support] error bars do not overlap**

diff btw mean amt of myelin are (possibly) significant / statistically significant / not due to chance;

3. **[support] diff mean % related to diff types of dementia;**

4. **[do not support] dementia may be due to other factors / not only due to lack of myelin (e.g. genetic)
as there are significant differences in myelin in different types of dementia;**

5. **[do not support] small sample sizes**

of $n < 25$ for some grps $\Rightarrow$ large error;

[3]

® do not support due to diff sample sizes

Cocaine is a drug that interferes with synaptic transfer. It affects neurones that release dopamine, an excitatory neurotransmitter associated with feelings of pleasure.

Fig. 5.2 shows the mechanism of action of cocaine.

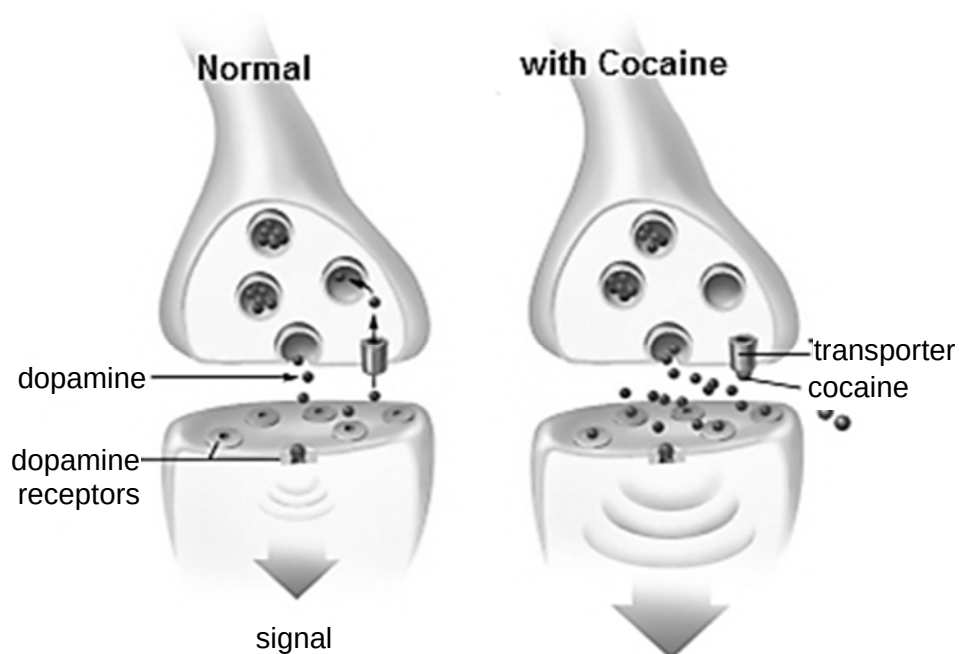


Fig. 5.2

- (c) (i) Describe the process of synaptic transfer in normal individuals.
1. **arrival of stimulus at synaptic knob ➡ opening of voltage-gated Ca^{2+} channel**

 Ca^{2+} influx;

 2. **translocation of synaptic vesicles containing dopamine to pre-synaptic mem (with aid of cytoskeleton)**

fuse pre-synaptic mem to release contents via exocytosis;

 3. **dopamine diffuses across synaptic cleft**

binds to ligand-gated Na^+ channels ➡ opening of ligand-gated Na^+ channel;

 4. **Na^+ influx ➡ depolarisation $\geq$ threshold pot of -50 mV**

generate action potential in post-synaptic neurone;
----- [4]
- (ii) With reference to Fig. 5.2, explain the effects of cocaine.
1. **cocaine blocks dopamine transporters ➡ ⊗ uptake of dopamine**

dopamine accumulates in the synaptic cleft;

 2. **con't binding of dopamine to dopamine receptor ➡ temporal summation (of graded potential)**

resulting in ↑ depol / opening of Na^+ gates thus continued feelings of pleasure;
----- [2]
- (iii) Long-term cocaine use results in a decrease in the number of dopamine receptors in the post-synaptic membrane.
- Suggest why cocaine abusers are no longer able to feel pleasure naturally.
- ↓ no. of receptors ➡ < dopamine can bind dopamine receptors**

➡ insufficient Na^+ influx to reach threshold pot thus fewer AP generated;
----- [1]

[Total: 13]

- 6 A region in Southern Australia used to have a huge lake system. About 500 000 years ago, the lakes started to dry up and they now consists of isolated small pools. Ten different species of a particular fish was found living in these pools. However, fossil evidence indicates that over 500 000 years ago there was only one species of the fish living in the lake.

(a) Explain how the ten species of fish evolved.

1. variation present

in ancestral pop of fish (due to spontaneous mutation);

2. drying up of land → diff env condition in each pool → diff selection pressures

indiv with favourable phenotype / adaptations selected for / have selective adv;

3. ↑ survival rate & repro success

advantageous / favourable alleles passed down to offspring → Δ in allelic freq in gene pools;

4. geographical isolation / geographical barrier of dry land

fish pop isolated i.e. prevent interbreeding / gene flow btw pops;

5. mutations arise independently in each pop

not shared btw gene pools;

6. ↑ genetic variation between pop → repro isolation / inability to interbreed

resulting in allopatric speciation / adaptive radiation;

[5]

- (b) Due to climate change, scientists predict that in 5000 years, water levels will rise and the pools will reform the huge lake system.

Suggest what might happen to the ten species of fish.

1. nos of each spp ↑ initially due to ↑ space / food;

**2. ↑ competition btw spp → reduction in nos for some / all spp
some spp outcompeted for food / habitat and therefore unable to survive / extinct;**

**3. spp may be able to interbreed due to lack of repro barriers
as they only diverged fr common ancestor recently (500 000 yrs ago);**

4. interbreeding btw spp → hybrids that ⊗ interbreed with parents → new spp;

5. diff spp occupies diff habitats / ecological niches;

**6. remain as diff spp / ⊗ interbreed due to presence of repro barriers
e.g. pre/post zygotic as gene pool has become too diversified;**

[2]

Tree frogs are commonly found in Southern Australia. *Hyla ewingi* and *Hyla verrauxi* are two closely related species of tree frogs. Research from breeding studies and DNA sequence data has shown that they have strong genetic compatibility.

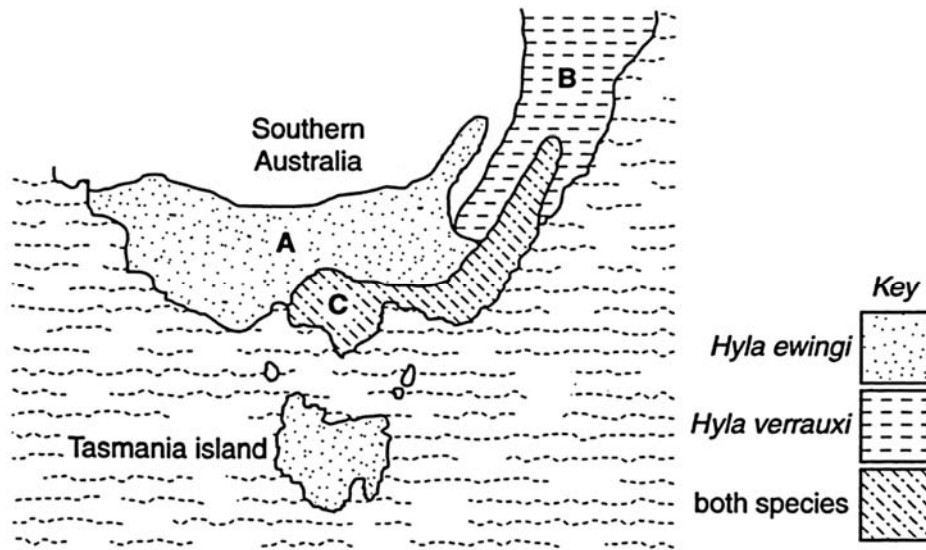


Fig 6.1

- (c) (i) State the genus of the tree frog.

Hyla;

[1]

- (ii) With reference to Fig. 6.1 and the research studies, explain how the scientists concluded that the two species of tree frogs are closely related.

1. **from breeding studies ➡ both spp able to interbreed**

to give viable fertile offspring;

2. **absence of genetic isolation / repro barrier**

gene flow btw spp still able to occur;

3. **strong genetic compatibility ➡ few diff btw DNA seq**

short time frame for accumulations of mutations / indicates recent divergence from common ancestor;

4. **present in same geographical region (of Southern Australia) overlapping habitat in C**

indicates close relatedness according to ESC / biogeographical evidence (as descendants from common ancestor tend to be located in the same region);

[4]

[Total: 12]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

- 7 **(a)** Describe the main stages in Calvin cycle. [8]

1. *occurs in stroma of chloroplast;*
2. *first stage: carbon fixation, where CO₂ combines with ribulose biphosphate;*
3. *catalysed by RuBP carboxylase / Rubisco;*
4. *forms unstable 6C intermediate, which breaks down to form 2 molecules of 3C glycerate-3-phosphate / PGA;*
5. *second stage: carbon reduction, where PGA phosphorylated → forms 1,3-bisphosphoglycerate using ATP;*
6. *then reduced to form 3C glyceraldehyde-3-phosphate / TP using reduced NADP;*
7. *1 out of 6 G3P exits cycle to be converted to glucose / carbo / lipids;*
8. *remaining 5 mol of G3P remains in cycle to regenerate (3 mols of) RuBP utilising ATP;*

- (b)** Describe the roles of membranes in photosynthesis and in respiration. [8]

1. *contains electron carriers of the ETC;*
2. *during e⁻ transfer, energy is released to pump H⁺ from mitochondrial matrix and stroma into mitochondrial intermembrane space and thylakoid space;*
3. *impermeability of membrane to ions;*
4. *allows formation of proton gradient across inner mito memb and thylakoid memb;*
5. *contains ATP synthase to harness proton motive force / which allows H⁺ to diffuse through;*
6. *to phosphorylate ADP to form ATP;*
7. *thylakoid membrane contains photosynthetic pigments that absorb light energy during photosynthesis;*
8. *membranes of mitochondrion contains transport proteins to allow transport of metabolites e.g. glucose / triose phosphate / pyruvate;*
9. *the envelopes of mito and chloroplast allows for compartmentalisation of the cell as mito. and chloroplast enz require different / specific conditions from the rest of the cell to function;*

(c) Explain how membrane fluidity is maintained in a cell when temperature increases. [4]

1. *increase temp ➡ increased KE ➡ phospholipids move further apart ➡ increased fluidity;*
2. *↑ cholesterol composition ➡ binds to FA tails of phospholipids to restrict movement ➡ ↓ memb fluidity;*
3. *↑ carbon chain length of FA tails ➡ ↑ SA for hydrophobic int ➡ ↓ memb fluidity;*
4. *↓ degree of saturation of FA tails ➡ straight chains ➡ more closely packed ➡ more hydrophobic int ➡ ↓ memb fluidity;*

8 (a) Describe the structural features of collagen and how they contribute to its function. [8]

1. *fibrous protein made up of amino acids linked by peptide bonds;*

[Struct to Fn] allows tight twisting to increase tensile strength

2. *tropocollagen, 3 polypeptides twisted into triple helix;*
3. *each polypeptide has gly-X-Y motif where X is usually proline (pro) and Y is usually proline or hydroxyproline (hyp);*
4. *3.3 a.a. per turn with gly in centre of helix, as it is the smallest amino acid ➡ allow tight twisting to ↑ tensile strength;*

[Struct to Fn] bonds / cross-linkages increase tensile strength

5. *presence of hydrogen bonds and covalent cross-links (at N-terminals) within the tropocollagen ➡ ↑ tensile strength;*
6. *covalent cross links form between N terminal and C terminal of tropocollagens involving Lys and Hyl ➡ ↑ tensile strength;*

[Struct to Fn] allows deposition of minerals to increase mechanical strength

7. *staggered arrangement / array of tropocollagen create hole region / 40nm gaps ➡ allow deposition of minerals ➡ ↑ mechanical strength;*

[Struct to Fn] bundling increases tensile strength

8. *collagen fibrils are further strengthened and stabilized by the formation of covalent cross-links involving Lys and Hyl between tropocollagen molecules in the fibril ➡ ↑ tensile strength;*
9. *bundling of collagen fibrils to form collagen fibres ➡ ↑ tensile strength;*
10. *Insolubility due to position of non-polar amino acids in exterior and interior of tropocollagen/ collagen fibres, do not allow interaction with water ➡ structural support;*

- (b) Insulin polypeptide synthesised by ribosomes attached to the endoplasmic reticulum will be transported out of the beta cell.

Describe the route taken by insulin out of the beta cell.

[8]

1. *insulin polypeptide / preproinsulin enters cisternal space of rER;*
2. *cleaving of signal peptide of preproinsulin ➡ formation of 3 disulfide bridges ➡ folding to form proinsulin;*
3. *further cleavage of C-peptide to form insulin protein, where chain A and B linked by 2 disulfide bridges;*
4. *transport vesicle containing insulin protein pinches off from rER and is transported to and fuses with membrane of cis face of GA;*
5. *protein is packaged, sorted, tagged in cisternae of GA;*
6. *addition, substitution and deletion of sugar monomers on oligosaccharide chain and phosphorylation to form molecular identification tag;*
7. *protein moves from cis face to trans face;*
8. *packaged into secretory vesicle & bud off from trans face of GA ➡ vesicle is transported to CSM via microtubules;*
9. *membrane of secretory vesicle fuses with CSM ➡ release of insulin out of beta cell via exocytosis;*

- (c) Explain the roles of vesicles that emerge from the Golgi body.

[4]

1. *secretory vesicles carrying proteins to be released out of the cell via exocytosis;*
2. *secretory vesicles containing memb components ➡ insert membrane components such as phospholipids / proteins into CSM;*
3. *lysosomes containing hydrolytic enz;*
4. *carry out phagocytosis of pathogens / food particles;*
5. *digestion of worn-out organelles / autophagy;*
6. *lysosomal pathway of apoptosis*
7. *AVP;*

[Total: 20]